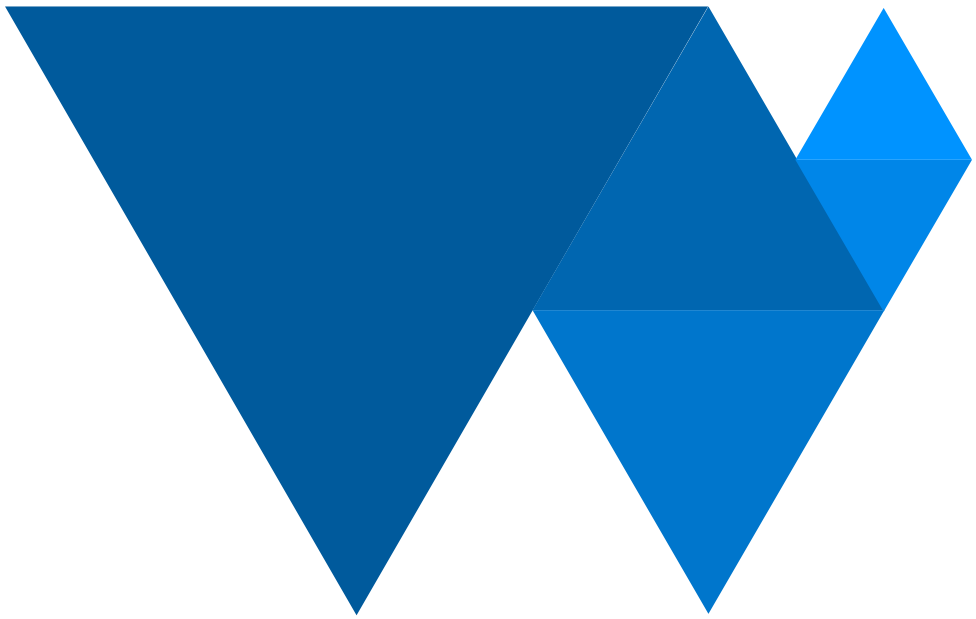




WebGPU in Firefox

Headlines

- Shipped on macOS (Apple Silicon)
- Linux looking good
- A year of security hardening
- More spec features
- Architectural improvements



macOS

- Firefox 145 for Tahoe, older versions in 147
- Apple Silicon only
- GPU process isolation



Linux

- Aiming for early 2027
- Presentation code: from proof-of-concept to shippable
- GPU process isolation



Security

- Many bounties paid
- Mozilla's AI-driven audits are working well
- When possible: secure by design
- Where are our blind spots?



Features

- dual-source-blending
- clip-distances
- float32-blendable
- primitive-index
- TRANSIENT_ATTACHMENT
- template list discovery
- subgroups (almost)
- … but gaps remain



Architecture

- wgpu-core-remote
- record at GPUEncoder finish
- deadlock detection
- "extension" API



Challenges

- Lazy initialization
- Buffer mapping races
- Limits
- Validating out native-only features



Questions?

